

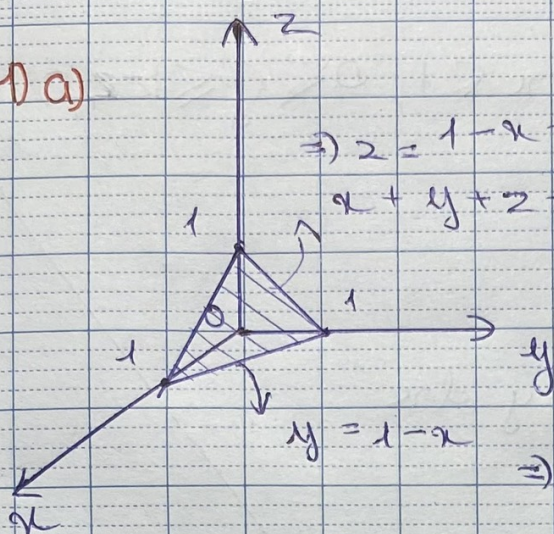
Tên: Nguyễn Thị Trà My

MSSV: 23127428

BÀI TẬP VỀ NHÀ TUẦN 6

Thứ.....ngày.....tháng.....năm.....

1a)



$$\Rightarrow z = 1 - x - y$$

$$x + y + z - 1 = 0$$

$$0 \leq x \leq 1$$

$$0 \leq y \leq 1 - x$$

$$0 \leq z \leq 1 - x - y$$

$$\Rightarrow T = \{(x, y, z) \mid 0 \leq x \leq 1, 0 \leq y \leq 1 - x, 0 \leq z \leq 1 - x - y\}$$

$$\iiint_T x^2 dV$$

$$= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} x^2 dz dy dx$$

$$= \int_0^1 \int_0^{1-x} x^2 z \Big|_{z=0}^{z=1-x-y} dy dx$$

$$= \int_0^1 \int_0^{1-x} x^2 (1-x-y) dy dx$$

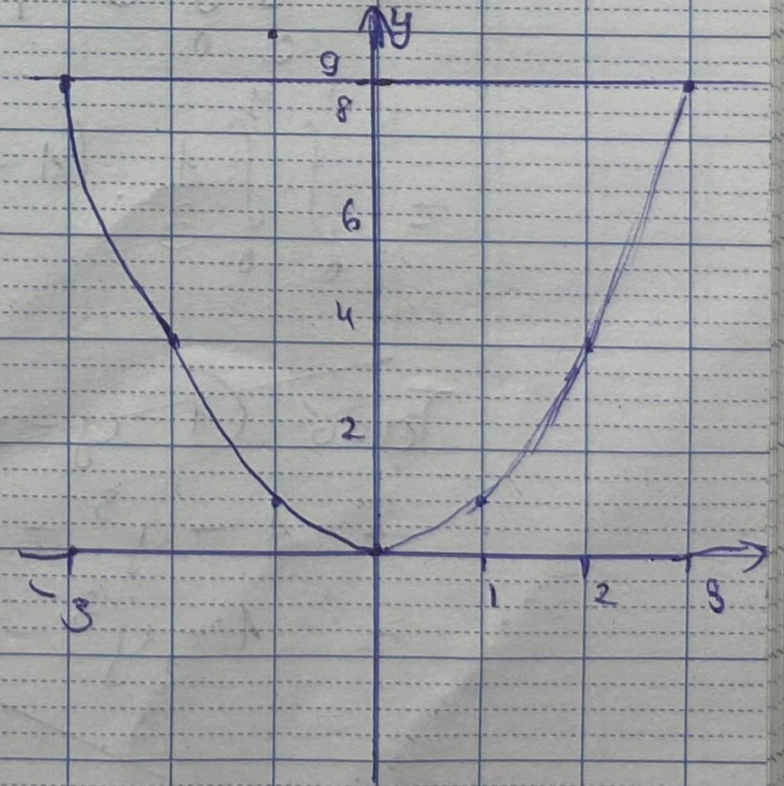
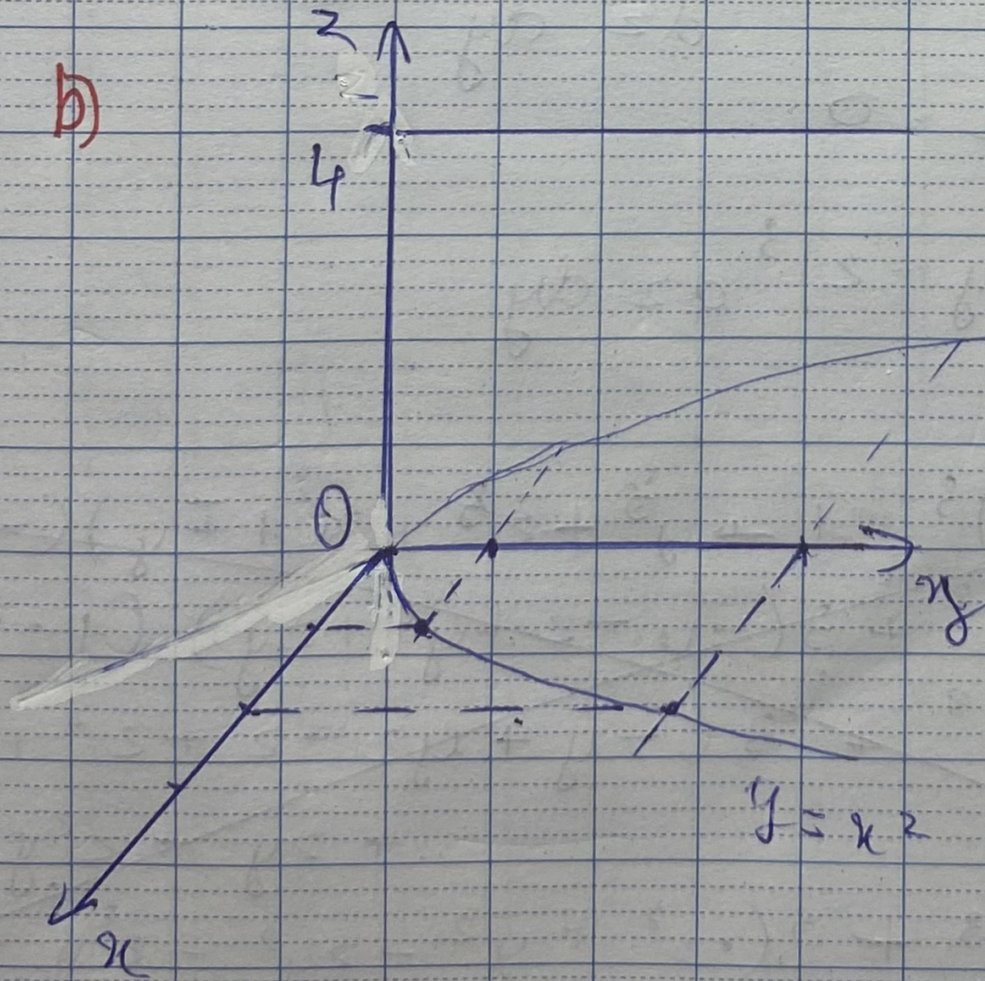
$$= \int_0^1 \int_0^{1-x} (x^2 - x^3 - x^2 y) dy dx$$

$$= \int_0^1 \left((x^2 - x^3) y - \frac{x^2 y^2}{2} \right) \Big|_{y=0}^{y=1-x} dx$$

$$= \int_0^1 \left((x^2 - x^3) (1-x) - \frac{(1-x)^2 x^2}{2} \right) dx$$

$$= \frac{1}{60}$$

b)



$$-8 \leq x \leq 8$$

$$0 \leq y \leq 9$$

$$0 \leq z \leq 4$$

$$-\sqrt{y} \leq x \leq \sqrt{y}$$

$$\Rightarrow I = \int (x, y, z) | -\sqrt{y} \leq x \leq \sqrt{y},$$

$$0 \leq y \leq 9, 0 \leq z \leq 4 \}$$

Thể tích của hình khối:

$$V = \int_0^4 \int_0^9 \int_{-\sqrt{y}}^{\sqrt{y}} dx dy dz$$

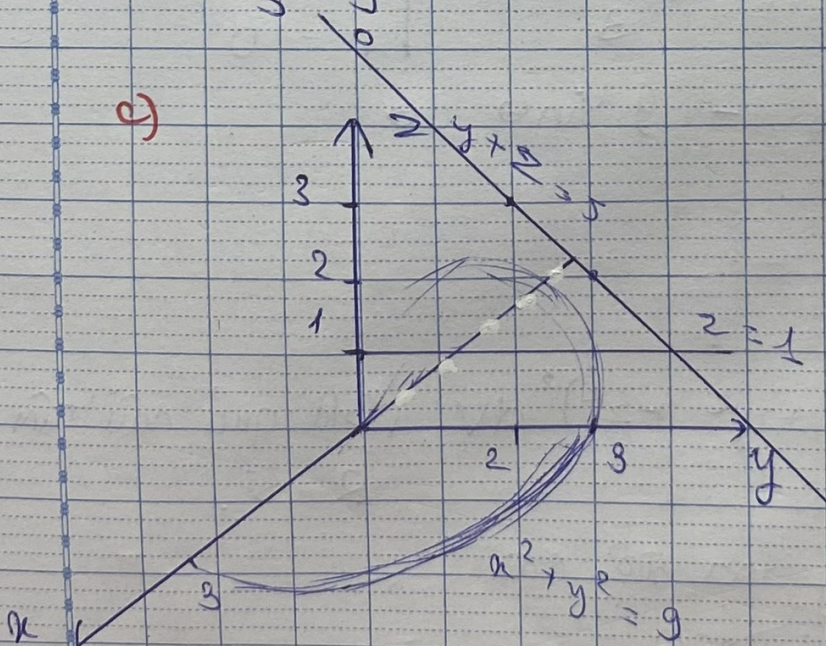
$$= \int_0^4 \int_0^9 x \Big|_{x=-\sqrt{y}}^{x=\sqrt{y}} dy dz$$

$$= \int_0^4 \int_0^9 2\sqrt{y} dy dz$$

$$= \int_0^4 \frac{4}{3} y^{3/2} \Big|_{y=0}^{y=9} dz$$

$$= \frac{4}{3} \int_0^4 27 dz = 144$$

g)



$$1 \leq z \leq 5 - y$$

Chuyển sang hệ tọa độ cực:

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\Rightarrow \begin{cases} 0 \leq \theta \leq 2\pi \\ 0 \leq r \leq 3 \end{cases}$$

Taco: $1 \leq z \leq 5-y$
 $\Rightarrow 1 \leq z \leq 5-r\sin\theta$

~~$\Rightarrow I = \int$~~

Thế tích của hình khối:

$$\begin{aligned}
 V &= \int_0^{2\pi} \int_0^3 \int_1^{5-r\sin\theta} r dz dr d\theta \\
 &= \int_0^{2\pi} \int_0^3 r z \Big|_{z=1}^{z=5-r\sin\theta} dr d\theta \\
 &= \int_0^{2\pi} \int_0^3 r(5-r\sin\theta - 1) dr d\theta \\
 &= \int_0^{2\pi} \int_0^3 r(4 - r\sin\theta) dr d\theta
 \end{aligned}$$

~~$= \int \frac{dr^2}{2}$~~

$$= \int_0^{2\pi} \left[2r^2 - \frac{r^3 \sin\theta}{3} \right]_{r=0}^{r=3} d\theta$$

$$= \int_0^{2\pi} (18 - 9\sin\theta) d\theta$$

$$= 36\pi$$

2)

a) $\iiint_B (x^2 + y^2 + z^2)^2 dV$, B là quả cầu tâm O, $R=5$.

Chuyển sang tọa độ cầu:

$$\begin{cases} x = \rho \sin \phi \cos \theta \\ y = \rho \sin \phi \sin \theta \\ z = \rho \cos \phi \end{cases} \quad 0 \leq \rho \leq 5, \quad 0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq \pi$$

Tacó:

$$\iiint (x^2 + y^2 + z^2)^2 dV$$

$$= \int_0^{2\pi} \int_0^\pi \int_0^5 \left[\rho^2 (\cos^2 \theta + \sin^2 \theta) + \rho^2 \cos^2 \phi \right] \rho^2 \sin \phi d\rho d\phi d\theta$$

$$= \int_0^{2\pi} \int_0^\pi \rho^4 \sin \phi d\rho d\phi d\theta$$

$$= \frac{1}{7} \int_0^{2\pi} \int_0^\pi \rho^7 \sin \phi \Big|_{\rho=0}^{\rho=5} d\phi d\theta$$

$$= \frac{1}{7} \int_0^{2\pi} \int_0^\pi 5^7 \sin \phi d\phi d\theta$$

$$= \frac{-5^7}{7} \int_0^{2\pi} \cos \phi \Big|_{\phi=0}^{\phi=\pi} d\theta$$

$$= \frac{-5^7}{7} \int_0^{2\pi} -2 d\theta = \frac{312500}{7} \pi$$

b)

$$\iiint_E e^{\sqrt{x^2 + y^2 + z^2}} dV$$

Tacó que centr. $\vec{r}^2 = x^2 + y^2 + z^2 = 9$

$$\Rightarrow \begin{cases} I = (0, 0, 0) \\ R = 3 \end{cases}$$

Chuyển sang tọa độ cầu.

$$\begin{cases} x = \rho \sin \phi \cos \theta \\ y = \rho \sin \phi \sin \theta \\ z = \rho \cos \phi \end{cases}$$

Tọa độ: $0 \leq \rho \leq 3$

$0 \leq \phi \leq \pi/2$

$0 \leq \theta \leq \pi/2$

$\Rightarrow E = \{(\rho, \phi, \theta) \mid 0 \leq \rho \leq 3, 0 \leq \phi \leq \pi/2, 0 \leq \theta \leq \pi/2 \}$

$$\iiint_E e^{\sqrt{x^2+y^2+z^2}} dV$$

$$= \int_0^{\pi/2} \int_0^{\pi/2} \int_0^3 e^{\rho} \rho^2 \sin \phi d\rho d\phi d\theta \quad (1)$$

Taxét: $\int_0^3 e^{\rho} \rho^2 \sin \phi d\rho$

$\begin{cases} u = \rho^2 \sin \phi \\ du = 2\rho \sin \phi d\rho \\ dv = e^{\rho} d\rho \\ v = e^{\rho} \end{cases} \Rightarrow$

$$\Rightarrow \int_0^3 e^{\rho} \rho^2 \sin \phi d\rho$$

$$= \rho^2 \sin \phi \cdot e^{\rho} \Big|_0^3 - 2 \int_0^3 e^{\rho} \rho \sin \phi d\rho \quad (2)$$

$\begin{cases} u' = \rho \sin \phi \\ du' = \sin \phi d\rho \\ v' = e^{\rho} \end{cases} \Rightarrow$

$$\Rightarrow 2 \int_0^3 e^{\rho} \rho \sin \phi d\rho = 2 \left[e^{\rho} \rho \sin \phi \Big|_0^3 - \int_0^3 e^{\rho} \sin \phi d\rho \right]$$

$$\begin{aligned}
 x &= 2(e^3 \cdot 3 \sin \phi - e^3 \sin \phi + \sin \phi) \\
 &= 2[\sin \phi (3e^3 - e^3 + 1)] \\
 &= 2 \sin \phi (2e^3 + 1)
 \end{aligned}$$

$$\begin{aligned}
 (2) \Rightarrow & \int_0^3 \int_0^{\pi/2} 2 \sin \phi (2e^3 + 1) d\phi d\theta \\
 &= 9 \sin \phi e^3 - 2 \sin \phi (2e^3 + 1) \\
 &= 9 \sin \phi e^3 - 4 \sin \phi e^3 - 2 \sin \phi \\
 &= 5 \sin \phi e^3 - 2 \sin \phi
 \end{aligned}$$

$$\begin{aligned}
 (1) \Rightarrow & \int_0^{\pi/2} \int_0^{\pi/2} 5 \sin \phi e^3 - 2 \sin \phi d\phi d\theta \\
 &= \int_0^{\pi/2} [-5 \cos \phi e^3 + 2 \cos \phi] \Big|_{\phi=0}^{\phi=\pi/2} d\theta \\
 &= \int_0^{\pi/2} 5e^3 - 2 d\theta \\
 &= 5e^3 \theta - 2\theta \Big|_0^{\pi/2} = 5e^3 \frac{\pi}{2} - \pi
 \end{aligned}$$

e)

Chuyển sang tọa độ cầu:

$$\begin{cases} x = \rho \sin \phi \cos \theta \\ y = \rho \sin \phi \sin \theta \\ z = \rho \cos \phi \end{cases}$$

$$\begin{aligned}
 \text{Thay đổi: } & 0 \leq \rho \leq 4 \cos \phi \\
 & 0 \leq \phi \leq \pi \\
 & 0 \leq \theta \leq 2\pi
 \end{aligned}$$

Thứ.....ngày.....tháng.....năm.....

$$\Rightarrow E = \rho(r, \phi, \theta) \mid 0 \leq \rho \leq 4 \cos \phi, 0 \leq \phi \leq \frac{\pi}{3}, 0 \leq \theta \leq 2\pi$$

$$V = \iiint dV$$

$$= \int_0^{2\pi} \int_0^{\pi/3} \int_0^{4 \cos \phi} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi/3} \left. \frac{\rho^3 \sin \phi}{3} \right|_{\rho=0}^{\rho=4 \cos \phi} d\phi \, d\theta$$

$$= \frac{1}{3} \int_0^{2\pi} \int_0^{\pi/3} 4^3 \cos^3 \phi \sin \phi \, d\phi \, d\theta$$

$$= \frac{4^3}{3} \int_0^{2\pi} \int_0^{\pi/3} \cos^3 \phi \sin \phi \, d\phi \, d\theta \quad (*)$$

Đặt $u = \cos \phi \Rightarrow du = -\sin \phi \, d\phi$

$$(*) \Rightarrow \frac{4^3}{3} \int_0^{2\pi} \int_1^{1/2} -u^3 \, du \, d\theta$$

$$= \frac{-4^3}{3} \int_0^{2\pi} \left. \frac{u^4}{4} \right|_{u=1}^{u=1/2} d\theta$$

$$= -\frac{4^2}{3} \int_0^{2\pi} -\frac{15}{16} \, d\theta = 10\pi$$